



Anticoagulant-associated bleeding: real-world data-driven pharmacovigilance and insights for bleeding management

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Background: Anticoagulant drugs are essential for managing thrombotic diseases but lead to increasing bleeding risk. The choice of suitable anticoagulant drugs requires an assessment of the patient's thrombotic risk factors and underlying ischemic pathology.

Methods: This study constructs a closed-loop framework integrating qualitative literature review and quantitative data analysis to evaluate bleeding risks in anticoagulant therapy. The qualitative review covers molecular mechanisms, risk factors, and nursing practices related to anticoagulant drugs. Quantitatively, we analyzed the bleeding events from the FDA Adverse Event Reporting System (FAERS) database (2004–2024), using disproportionality analysis with Reporting Odds Ratio, Proportional Reporting Ratio, Bayesian Confidence Propagation Neural Network, and Empirical Bayes Geometric Mean to identify significant bleeding signals. For each anticoagulant drug, the 10 strongest signals were extracted and anatomically profiled to construct a drug-specific hemorrhage atlas. Logistic regression analyses were applied to investigate the effects of age, sex, reporter, and comorbidities on the occurrence of anticoagulant drugs-associated bleeding. To address the critical issue of bleeding risk management in the clinic, we further discussed the multidisciplinary nursing across the anticoagulation therapy cycle.

Results: The study revealed significant variations in bleeding risks among different anticoagulants. Anticoagulant-associated bleeding comprised 110 701 reports from FAERS database. Elderly patients (≥65 years old) supplied 63.2% of cases; 32% of events emerged within 30 days of therapy initiation. Hospitalization was documented in 46.8% and death in 20.7%. Spinal subdural hemorrhage yielded the highest signal (ROR 144.79, 95% CI 28.09–746.33), followed by adrenal, mesenteric, and intracranial hemorrhages among all anticoagulant drugs. Gastric/lower gastrointestinal bleeding appeared in the top-10 list of every oral anticoagulant, while retroperitoneal bleeding was a shared high-signal event among heparins and parenteral direct thrombin inhibitors. Betrixaban exhibited extreme RORs for traumatic (ROR 18 059.67, 95% CI: 8829.21–36 940.05) and intracranial (ROR 643.97, 95% CI: 249.82–1660) bleeding. Each drug also exhibited distinct bleeding characteristics and differences in organ system distribution. Multivariable logistic regression across four hierarchical models identified agents with distinct, model-independent bleeding risk profiles relative to warfarin.

Conclusions: This large-scale pharmacovigilance study provides the most comprehensive, publicly available atlas of anticoagulant-related bleeding signals. The pronounced inter-drug heterogeneity and persistent long-term risk mandate individualized choice of agent and prolonged monitoring to ultimately improve the safety–efficacy balance of antithrombotic therapy.

Keywords: anticoagulant, bleeding, management, nurse, safety

Introduction

Anticoagulant drugs are crucial for thrombotic disease management and widely used in conditions like venous thromboembolism (VTE), atrial fibrillation (AF), and valvular heart disease^[1]. The evolution from traditional heparin and warfarin to novel oral anticoagulants (NOACs) has enhanced efficacy and safety in the emerging anticoagulant field^[2,3]. For instance, the use of

apixaban and edoxaban demonstrated a greater net clinical benefit compared to vitamin K antagonists (VKAs) in stroke prevention among elderly patients with AF^[3]. However, all of the anticoagulation therapy has a dual-edged nature, increasing bleeding risk, a major concern in such treatment^[4]. Bleeding risks vary among anticoagulants^[3,5]. Patient differences, comorbidities, and drug interactions further complicate bleeding

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prevention and management^[6]. The choice of suitable anticoagulant drugs should be guided by an assessment of the patient's thrombotic risk factors and underlying ischemic pathology^[7]. It has shown that clinicians place more importance to bleeding risk, and the anticoagulant prescribing is largely influenced by bleeding risk than stroke risk^[8]. Currently, the safety challenges of anticoagulant drugs are increasingly prominent, and the bleeding risk of anticoagulant therapy is a clinical issue that requires high attention.

Numerous studies, while examining the efficacy of anticoagulant drugs, also pay close attention to the associated bleeding risks^[9,10]. A meta-analysis of 31 prospective and retrospective studies (including 287 692 patients) reported that there was no significant difference in the risk of major gastrointestinal bleeding between oral anticoagulants (OACs) or low molecular weight heparin, but factor Xa inhibitors were associated with a lower risk of all severities of gastrointestinal bleeding compared with warfarin or dabigatran in the secondary analysis^[11]. Another meta-analysis involving 440 281 patients ≥ 75 years of age with AF showed that NOAC was associated with a lower risk of systemic embolism, intracranial hemorrhage, hemorrhagic stroke, and fatal bleeding compared with warfarin, but with an increased risk of gastrointestinal bleeding^[12]. Additionally, the development of anticoagulant drugs with minimal bleeding risk has become a key research focus. Striking a balance between the risk of thrombosis and bleeding remains a critical challenge that must be addressed (Fig. 1). Multidisciplinary collaboration is key in anticoagulation management. As a vital component of the healthcare team, nurses play a vital role in patient education, adherence management, and early bleeding detection, ensuring patient safety^[13,14]. It suggests that the close collaboration among doctors and nurses enables comprehensive patient assessment and personalized anticoagulation strategies.

Despite the extensive research, there are still gaps in the comprehensive understanding of bleeding events associated with different anticoagulants in real-world settings and the role of nursing care in managing these risks. This study aims to integrate a targeted literature review with the FDA Adverse Event (AE) Reporting System (FAERS) data mining to dissect anticoagulant-related bleeding, benchmark drug-specific risks, and outline nurse-led, closed-loop risk management. The closed loop in this work was completed by (1) generating pharmacovigilance signals through four-algorithm disproportionality analysis of FAERS data, (2) synthesizing evidence on mechanisms and risk factors linked to high-signal events, and (3) translating the integrated findings into actionable nurse-associated risk-management strategies that feed

HIGHLIGHTS

- Balancing thrombotic and bleeding risks with anticoagulant therapy remains a critical challenge.
- First confounder-adjusted bleeding atlas for 12 marketed anticoagulants was analyzed.
- 46.8% were hospitalized and 20.7% died in patients with anticoagulation-related bleeding.
- Spinal subdural hemorrhage yielded the highest signal, followed by adrenal, mesenteric, and intracranial hemorrhages.
- Multidisciplinary strategies would enhance the safety and efficacy of anticoagulation therapy.

back into routine surveillance and patient education, aiming to creating a continuous quality-improvement cycle.

Methods

Qualitative literature review of bleeding risk with anticoagulant drug treatment

This study adopted a mixed-method design, combining quantitative data analysis and qualitative literature review, to comprehensively evaluate the bleeding risk and management strategies in anticoagulant therapy. The data of qualitative literature reviews come from commonly used databases such as PubMed, Web of Science, China National Knowledge Infrastructure, and Wanfang Data. The search time is up to April 2025. The literature related to anticoagulant drug treatment, bleeding risk, and nurses' participation in management was included, such as the observational studies, randomized controlled trial, meta-analysis, and guideline consensus. The search keywords include anticoagulant therapy, anticoagulant drug, warfarin, NOACs, rivaroxaban, dabigatran, apixaban, edoxaban, bleeding risk, hemorrhage risk, bleeding complications, bleeding assessment, and nurses.

Quantitative analysis of bleeding events with anticoagulant drug treatment

Data sources and extraction

The quantitative data analysis for this study was derived from the FAERS database, spanning AE reports from the first quarter of 2004 to the third quarter of 2024. FAERS serves as

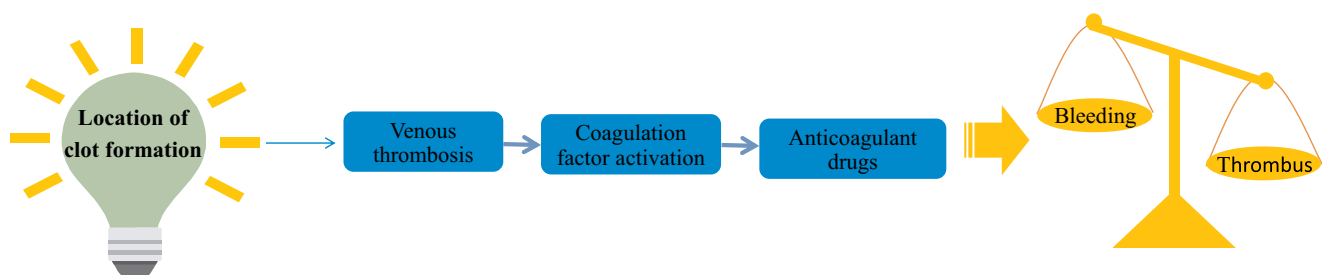


Figure 1. Location of thrombus formation and anticoagulant strategies.

a comprehensive, publicly available repository of self-reported AEs, collated from healthcare professionals, pharmaceutical manufacturers, and patients across multiple countries. For this analysis, reports of AEs associated with anticoagulant drugs were retrieved by querying the Standardized MedDRA Queries files. The anticoagulant drugs included warfarin, rivaroxaban, apixaban, edoxaban, betrixaban, dabigatran, heparin, enoxaparin, dalteparin, fondaparinux, bivalirudin, and argatroban. Each American Standard Code for Information Interchange file is composed of seven distinct sections: patient demographics and administration information, drug details, AEs, patient outcomes, report sources, start and end dates of drug therapy, and indications for use or diagnosis.

In the FAERS database architecture, the aforementioned files are linked via unique identifiers such as PRIMARYID. To ensure the completeness and accuracy of the data, the following measures were taken. Data cleaning and standardization: The Medex_UIMA_1.3.8 software was employed to standardize and unify the drug names within the database. This process helps in consolidating data entries and reducing discrepancies. Duplicate removal: A two-step deduplication process was implemented. Initially, the data were normalized and cleaned to remove any duplicate rows. Subsequently, deduplication was performed based on the latest FDA_DT when both CASEID and FDA_DT were identical. Data validation: We cross-referenced the extracted data with the original reports to ensure accuracy. This involved verifying the consistency of drug names, AE terms, and patient demographic information. Handling missing data: For reports with missing data (e.g., incomplete patient information), we only included reports where the anticoagulant drug was clearly identified as the primary suspect drug. Reports with significant missing data that could affect the analysis were excluded. These measures ensure the integrity and accuracy of the data extracted from the FAERS database, laying a solid foundation for subsequent analysis. Data extraction was conducted based on the drug name of each anticoagulant drug with the requirement. AE signals were classified and described using the preferred term (PT) from MedDRA version 25.1.

Disproportionality analysis

Disproportionality analysis, a cornerstone pharmacovigilance technique in the FAERS database, evaluates the association between a drug and an AE by comparing the disproportionality ratio of the target drug and AE to that of other drug–AE pairs. Rooted in distinct statistical principles, this method can be categorized into frequency-based and Bayesian approaches. Both quantify disproportionality through a 2×2 contingency table (Supplemental Digital Content Table S1, available at: <http://links.lww.com/JS9/G314>) to identify potential pharmacovigilance risk signals. However, neither approach can be deemed the gold standard, as each possesses unique strengths and limitations. The frequency-based method is highly sensitive but less specific and may be influenced by individual values, while the Bayesian method is more specific and stable, albeit generally less sensitive.

In this study, we employed four diverse algorithms to detect AE signals: the frequency-based Reporting Odds Ratio (ROR) and Proportional Reporting Ratio (PRR), as well as the Bayesian Confidence Propagation Neural Network (BCPNN) and

Empirical Bayes Geometric Mean (EBGM) methods. ROR and PRR are frequentist measures with high sensitivity suitable for initial signal screening, whereas BCPNN and EBGM are Bayesian shrinkage estimators that enhance specificity and stability in large, heterogeneous databases such as FAERS. Detailed formulas and threshold values for these algorithms are provided in Supplemental Digital Content Table S2, available at: <http://links.lww.com/JS9/G314>. For the frequency-based methods, a signal was considered positive if the lower limit of the 95% confidence interval (CI) of ROR was greater than 1, and the number of reports containing both the suspect drug and AE (a) was ≥ 3 . For PRR, a signal was deemed positive if $PRR \geq 2$, $\chi^2 \geq 4$, and $a \geq 3$. For the Bayesian methods, a signal was considered positive if $EBGM_{05} > 2$ and if the lower limit of the 95% CI of the Information Component (IC025) was greater than 0 for BCPNN. A higher value for each metric indicated a stronger signal, suggesting a more robust correlation between the target drug and AE. In this study, a signal was confirmed only when detected by all four algorithms concurrently.

The work has been reported in line with the Standards for Quality Improvement Reporting Excellence (SQUIRE) criteria^[15]. The work has been reported in line with the TITAN criteria^[16]. The research team employed a rigorous methodology to ensure that the findings were of high quality, reliable, and reproducible.

Ethics and study approval

This study utilized data from the FAERS database, a publicly available and anonymized resource designed to support the FDA-approved drug safety surveillance program and accessible to the public for research purposes. Data extraction and use were conducted in full compliance with the FAERS Data Use Agreement, including the FDA's Terms of Use for research purposes. Given the anonymized nature of the data and its intended use for surveillance and research, Institutional Review Board approval, as well as patient consent, were not required.

Statistical analysis

Demographic characteristics (age, sex, and weight), administrative details (report source and country), time-to-onset, and patient outcomes were extracted from each AE report. Time-to-onset was defined as the interval between the start of therapy and the occurrence of the AE, measured in days. Logistic regression analyses were applied to investigate the effects of age, sex, reporter, and comorbidities (hypertension and gastric ulcer) on the occurrence of anticoagulant drugs-associated bleeding. Data processing and analysis were conducted using R version 4.2.3 (R Foundation for Statistical Computing, Vienna, Austria).

Results

Qualitative literature review of bleeding risk with anticoagulant drugs

Indications of anticoagulant drugs and influencing factors of bleeding

Anticoagulant drugs inhibit the clotting process by affecting the activity or amount of different clotting factors, thereby preventing the formation of blood clots (Fig. 2)^[17]. Anticoagulant drugs mainly include heparin and OAC, the latter divided into NOAC

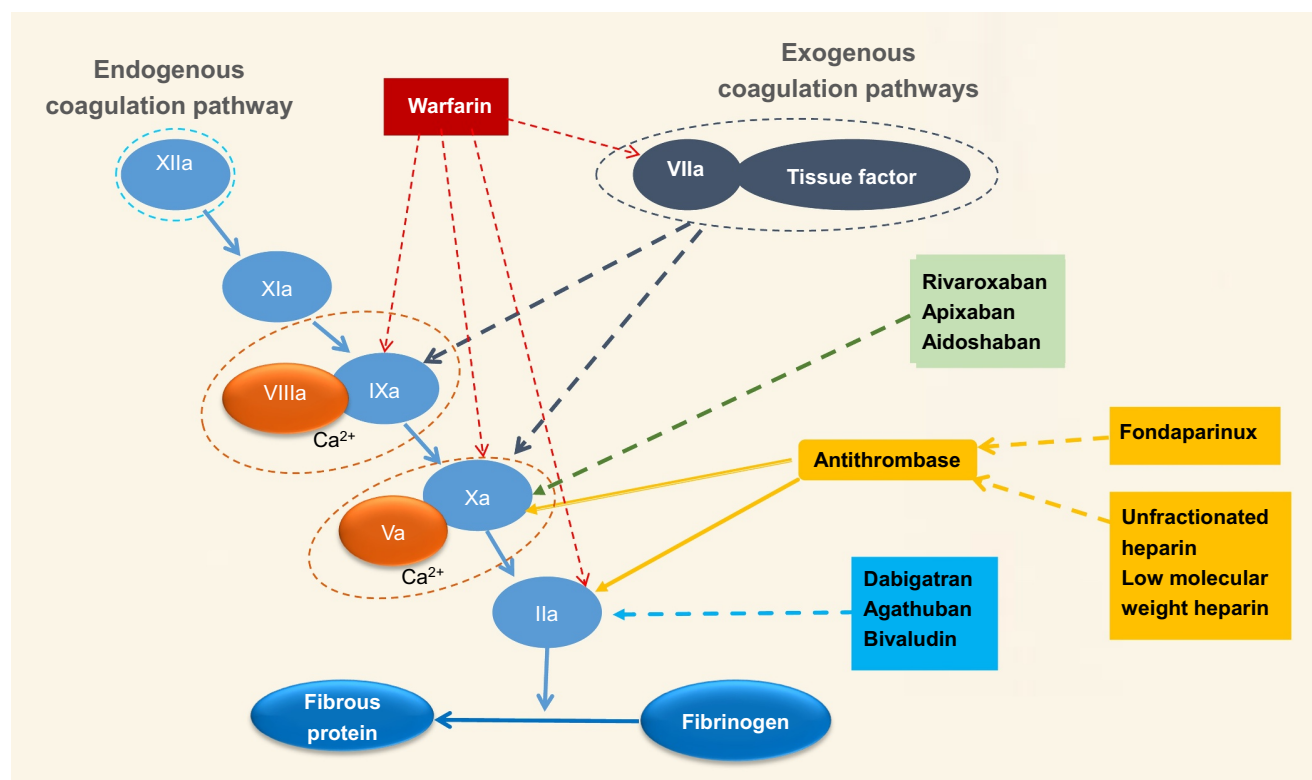


Figure 2. Mechanism of anticoagulant drugs in preventing and treating thrombosis. The dashed arrows in the figure indicate inhibition of the corresponding targets.

(dabigatran targeting thrombin and factor Xa targeting apoxaban, betrixaban, edoxaban, and rivaroxaban) and the VKA warfarin (Fig. 3)^[18]. Anticoagulant therapy is preferred for AF, VTE, and valvular heart disease, while the most common concern is also bleeding^[19]. NOAC is often as effective as warfarin but is less susceptible to drugs, food, and bleeding risk and does not require regular laboratory monitoring^[20]. Patients with deep venous thrombosis (DVT) and pulmonary embolism (PE) often

require 3 months of anticoagulant therapy with heparin, warfarin, or NOAC, after which treatment is based on the risk of VTE recurrence, such as transient risk factors, unprovoked or malignancy-associated, and the risk of treatment-induced major bleeding^[21]. NOAC has largely replaced warfarin-based VKAs as the main indication for oral anticoagulation, prevention and treatment of VTE, and prevention of embolic stroke in AF, but there may be specific limitations or contraindications such as

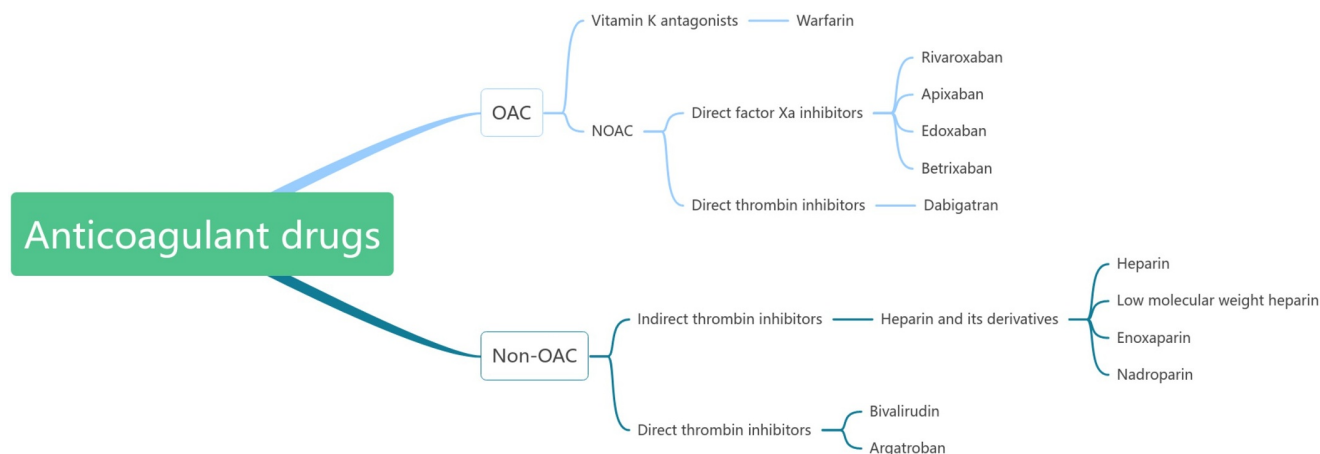


Figure 3. Classification of anticoagulant drugs. NOAC, novel oral anticoagulant; OAC, oral anticoagulant.

insufficient effectiveness in high thrombotic risk conditions including antiphospholipid syndrome and mechanical heart valves. Patients with cancer-related thrombosis may benefit from NOAC, but the risk of bleeding must be weighed in those with gastrointestinal or genitourinary tumors and those who are weak, overweight, or have moderate to severe chronic kidney disease, which may require laboratory monitoring^[22]. In many cases, bleeding caused by anticoagulant drugs is very insidious (such as a small amount of gastrointestinal bleeding), which is not easy to detect. In some cases, the use of anticoagulant drugs will also appear as black stool, ecchymosis, and even lead to excessive bleeding. The risk of bleeding is always present if anticoagulant drugs are used. Recent real-world reports in children showed that the incidence of major bleeding of NOAC per year was 0–4.9 per 1000 persons, while that of warfarin was 4.3–6.7 per 1000 persons, and heparin was 3.7–4.6 per 1000 persons^[23].

Bleeding becomes a common and potentially fatal complication during anticoagulant therapy. It is necessary to continue taking anticoagulant drugs and monitoring bleeding under most conditions. For moderate bleeding (e.g., decreased hemoglobin > 30 g/L), reduction of anticoagulant dose may be considered. For severe bleeding (e.g., hemoglobin reduction > 50 g/L and gastrointestinal bleeding), discontinuing OACs and using activated carbon adsorption and hemodialysis should be considered. The methods of hemostasis include pressure hemostasis, nasal tamponade, and emergency endoscopy, as well as reduction of the impact of blood loss (such as oxygen inhalation, intravenous fluids, and blood transfusions). In fact, the specific reversal agents, antifibrinolytic agents, or prothrombin complex concentrates are the most critical measures^[24]. A meta-analysis showed that warfarin, dipyridamole, non-acetylsalicylic non-steroidal anti-inflammatory drugs (NSAIDs), selective cyclooxygenase-2 inhibitors, and glucocorticoids were positively correlated with the incidence of aneurysmal subarachnoid hemorrhage^[25]. Antihypertensive drugs, statins, biguanides, and acetylsalicylic acid were inversely associated with the risk of aneurysmal subarachnoid hemorrhage and were candidates for its prevention^[25]. Patients with intracranial aneurysms should carefully determine the indications for these drugs.

A proportion of NOAC apixaban (~15% of an oral dose), edoxaban (<10%), and rivaroxaban (~18%) were cleared by CYP3A4/5, and the bioavailability and clearance of betrixaban, edoxaban, and rivaroxaban could be increased by P-glycoprotein (P-gp) inhibitors^[26]. It was reported that patients with nonvalvular AF (a common indication of NOAC) tend to be older and have more comorbidities, with multidrug use often reaching 40–77%, and then leading to adverse NOAC interaction outcomes such as bleeding^[26]. It showed that the combination of CYP3A4 and P-gp inhibitors could increase rivaroxaban exposure in patients with renal damage, but there was no dose relationship^[26]. Compared with enoxaparin, patients with creatinine clearance <30 mL/min had approximately double betrixaban exposure and a significantly increased risk of bleeding^[26]. Besides, taking NOAC together with an antiplatelet drug (acetylsalicylic acid or clopidogrel) or a NSAID (naproxen) increased the risk of bleeding because bleeding duration is greatly prolonged^[26]. In patients with first non-induced VTE receiving prolonged anticoagulation therapy (warfarin or NOAC) for more than 3–6 months, the incidence of major bleeding was significantly higher in patients older than 65 years or with creatinine clearance below 50 mL/

min, a history of bleeding, co-use of antiplatelet drugs, or hemoglobin levels below 100 g/L^[27]. The case fatality rate for warfarin was 8.3 and 9.7% for NOAC^[27]. Treatment with warfarin in combination with antiplatelet drugs, many antimicrobials, NSAIDs, selective serotonin reuptake inhibitors, mirtazapine, and loop diuretics increased the risk of drug-related bleeding, while proton pump inhibitors were associated with reduced hospitalizations for upper gastrointestinal bleeding in warfarin patients^[28]. No significant effect was found on thromboembolic events or mortality in combination with warfarin^[28].

Evidence-based comparison of anticoagulant regimens and bleeding risk

A meta-analysis of 31 prospective and retrospective studies (including 287 692 patients) reported the risk of gastrointestinal bleeding and found that there was no significant difference in major gastrointestinal bleeding risk with OAC or low molecular weight heparin, but factor Xa inhibitors were associated with a lower risk of all severities of gastrointestinal bleeding compared with warfarin or dabigatran in the secondary analysis^[29]. In 22 studies involving 440 281 patients ≥ 75 years of age with AF, NOAC was associated with a lower risk of systemic embolism, intracranial hemorrhage, hemorrhagic stroke, and fatal bleeding compared with warfarin, but with an increased risk of gastrointestinal bleeding^[12]. Rivaroxaban was associated with a higher risk of major bleeding than apixaban and edoxaban, while that of dabigatran was higher than apixaban^[12]. In patients with left ventricular thrombus, there was no significant difference between warfarin and NOAC treatment for stroke, systemic embolism, or thrombosis elimination, but the latter had lower all-cause mortality and bleeding events^[30]. Another meta-analysis of 28 studies involving 74 660 patients using bioartificial heart valves showed that NOAC reduced the risk of all-cause bleeding, stroke, systemic embolism, and intracranial hemorrhage outcomes compared with warfarin, but there was no significant difference in major bleeding and all-cause mortality between them^[31]. Compared with antiplatelet therapy, patients taking OAC during oral implant surgery had an increased risk of bleeding, and warfarin users may have a higher tendency to bleed than NOAC patients, whereas continued treatment with OAC did not increase the risk of bleeding if local hemostasis is used^[32]. A systematic review of extracorporeal membrane oxygenation without anticoagulants showed that the incidence was comparable in circuitry and patient thrombosis compared with patients receiving continuous systemic anticoagulation combined with extracorporeal membrane oxygenation^[33].

In a meta-analysis of 11 studies involving 5039 patients with carotid artery dissection (30% anticoagulant and 70% antiplatelet), the risk of death and symptomatic intracranial hemorrhage was similar for anticoagulant and antiplatelet therapy, but the risk of ischemic stroke was lower for anticoagulants, and the risk of major bleeding was higher^[34]. It has been shown that continuous anticoagulation therapy after gastrointestinal bleeding has a lower chance of thromboembolic events, death, and increased incidence of gastrointestinal bleeding than those who stop treatment, and thus continuous anticoagulation therapy is recommended^[35]. For the long-term treatment of DVT, NOAC was virtually no different than warfarin or heparin in preventing VTE recurrence, DVT recurrence, PE, and all-cause mortality,

but NOAC reduced the rate of major bleeding^[36]. There was also a work that revealed that the risks of VTE recurrence, major hemorrhage, intracranial hemorrhage, death, and complete venous recanalization were comparable between NOAC and warfarin in patients with cerebral venous thrombosis, indicating their considerable efficacy and safety^[37]. In patients with anti-phospholipid syndrome, NOAC use increased the odds of subsequent arterial thrombotic events, particularly stroke, recurrence of arterial thrombotic events, or VTE compared with warfarin, but there was no significant difference in their odds of developing VTE or major bleeding^[38]. Apixaban and rivaroxaban were available as options for prolonged prevention of blood clots after cancer surgery (weak recommendation), but high-quality evidence just recommended apixaban as an option for VTE treatment^[39]. To evaluate the benefits and harms of different anticoagulants for the prevention of VTE in acute hospitalized patients, evidence from 44 randomized controlled trials involving 9095 participants showed that pentasaccharid, medium-dose low-molecular-weight heparin, NOAC, and intermediate-dose unfractionated heparin were the most likely to reduce symptomatic venous thrombosis^[40].

However, none of the anticoagulants reduced all-cause mortality, and direct OACs and intermediate-dose unfractionated heparin were most likely to increase major bleeding^[40]. These reports highlighted the evaluation for the net clinical benefit and bleeding risk in different crowds to determine the prior anticoagulant strategy.

The necessity of real-world data across the world

The above qualitative literature review provides valuable insights into the molecular mechanisms, risk factors, and clinical trials related to bleeding events associated with anticoagulant drugs and anticoagulant scheme. However, this approach has several limitations that restrict its comprehensiveness and applicability. First, many studies included in the review may have excluded high-risk populations, such as those with severe comorbidities or specific contraindications, leading to an incomplete understanding of bleeding risks in these vulnerable groups. Second, the sample sizes of the studies reviewed are often limited, which may reduce the statistical power and generalizability of the findings. Third, the follow-up periods in these studies are frequently short, making it difficult to assess the long-term bleeding risks associated with anticoagulant therapy. These limitations highlight the necessity of complementing the qualitative literature review with real-world data, which can provide a more comprehensive and dynamic picture of bleeding risks across a broader population, including high-risk patients, and over longer periods of observation. This integration of real-world evidence is essential for developing more robust and personalized anticoagulation management strategies that address the diverse needs of patients in clinical practice.

Real-world bleeding data analysis from the FAERS database

Characteristics of bleeding event reports for anticoagulant drugs

This study comprehensively analyzed 110 701 AE reports associated with anticoagulant drugs from the FAERS database spanning from 2004 to 2024. The data revealed several key demographic and clinical characteristics among the reported

Table 1

Characteristics of AE reports related to anticoagulant drugs from the FAERS database

Items	Number of reports (%)
Sex	
F	46 380 (41.9%)
M	52 391 (47.3%)
Missing	11 930 (10.8%)
Weight	
<50 kg	1788 (1.6%)
>100 kg	6088 (5.5%)
50–100 kg	24 173 (21.8%)
Missing	78 652 (71.0%)
Age	
<18	272 (0.2%)
18–64	18 780 (17.0%)
65–85	51 589 (46.6%)
>85	10 594 (9.6%)
Missing	29 466 (26.6%)
Reporters	
Consumer	49 055 (44.3%)
Health-professional	7320 (6.6%)
Lawyer	732 (0.7%)
Medical doctor	29 057 (26.2%)
Other health-professional	8645 (7.8%)
Pharmacist	14 274 (12.9%)
Registered nurse	13 (0.0%)
Missing	1605 (1.4%)
Country of the reporter (top 10)	
United States	81 399 (73.5%)
Japan	3982 (3.6%)
Germany	3747 (3.4%)
France	2899 (2.6%)
United Kingdom	2673 (2.4%)
Italy	2143 (1.9%)
Canada	1771 (1.6%)
Australia	748 (0.7%)
China	692 (0.6%)
Brazil	483 (0.4%)

AE, adverse event; FAERS, FDA Adverse Event Reporting System.

cases (Table 1). In terms of gender distribution, male patients accounted for 47.3% of the bleeding-related AE reports, while female patients made up 41.9%. The weight data, though largely incomplete, showed that 21.8% of the patients were in the 50–100 kg range, 5.5% were over 100 kg, and 1.6% were under 50 kg. The age distribution indicated a significant predominance of elderly patients, with 46.6% of reports involving patients aged 65–85, followed by 17.0% for those aged 18–64, 9.6% over 85, and a minimal 0.2% under 18.

Consumers were the primary reporters, contributing 44.3% of the cases, followed by medical doctors (26.2%), while registered nurses just reported 13 cases. Geographically, the reports were submitted from 170 countries, with the United States being the most predominant source (73.5%), followed by Japan (3.6%), Germany (3.4%), and France (2.6%).

Time-to-onset of bleeding AE

The temporal pattern of AE onset following anticoagulant drug use exhibited substantial variability (Fig. 4A). The findings showed that a substantial proportion of bleeding AE occurred

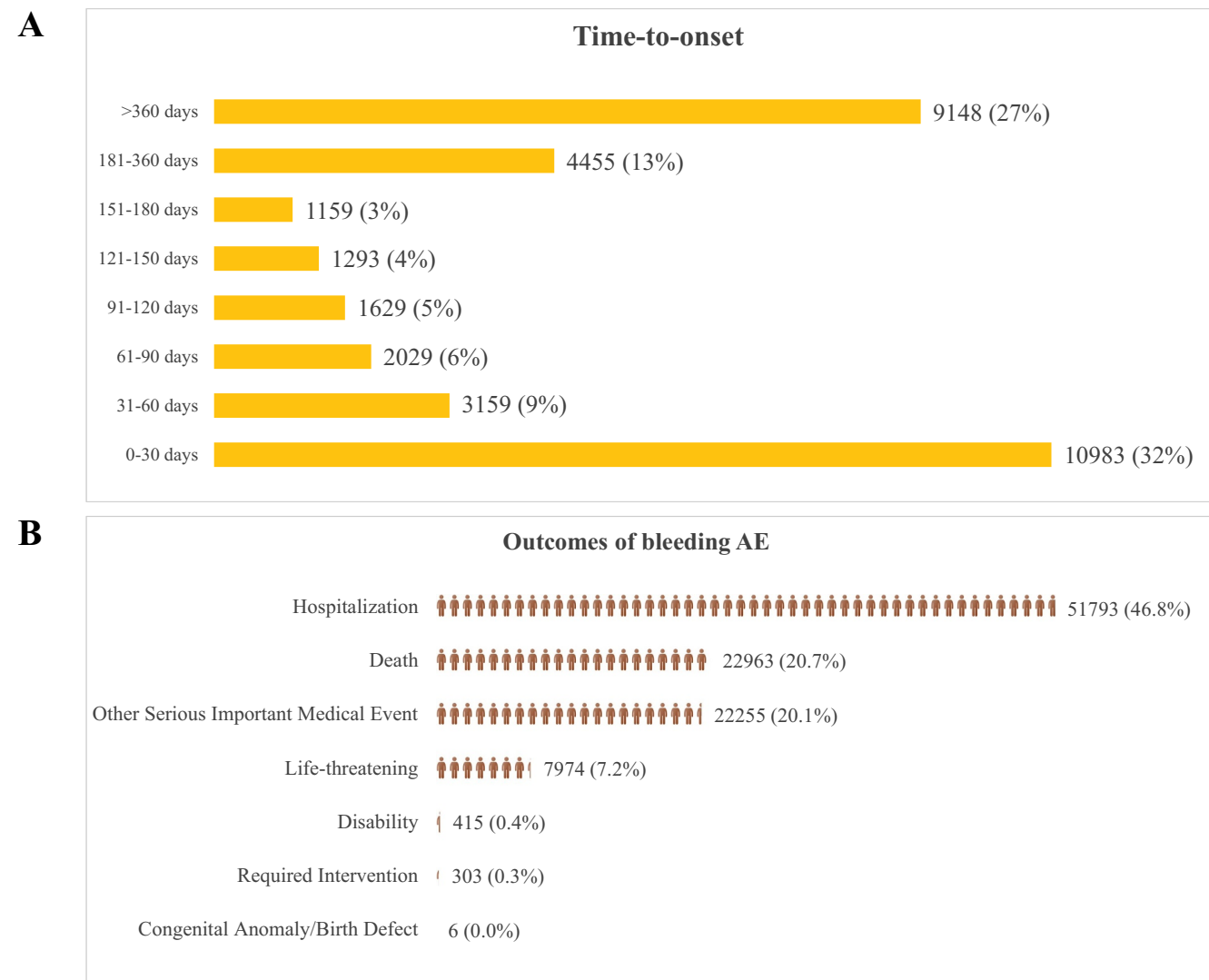


Figure 4. The cases and ratio of different time-to-onset (A) and clinical outcomes (B) of bleeding AE. AE, adverse event.

within the initial 30 days of anticoagulant therapy, accounting for 32% of the reported cases. This suggests that the early phase of treatment may pose a heightened risk for bleeding complications. However, a notable number of bleeding events also emerged beyond this initial period, with 27% occurring after 360 days of treatment. This extended temporal distribution highlights the importance of continuous monitoring and risk assessment throughout the entire course of anticoagulant therapy, emphasizing that the risk of bleeding is not confined to the early stages but persists over the long term.

Clinical outcomes of bleeding AE

The adverse outcomes related to bleeding events associated with anticoagulant drugs also revealed a diverse range of clinical consequences (Fig. 4B). The clinical outcomes of the bleeding AE were predominantly severe. It indicated that hospitalization is the most common adverse outcome, accounting for 46.8% of the reports. Mortality is a significant concern, representing 20.7% of the cases. Other serious medical events, including

life-threatening conditions, make up 20.1% of the reports. Notably, life-threatening events alone account for 7.2% of the total, while disability and required intervention are relatively less frequent at 0.4% and 0.3%, respectively. Though the missing data correspond to 4992 (4.5%) cases, these findings highlight the risk of anticoagulant therapy with severe bleeding complications.

Signal detection at the PT level of all anticoagulant drugs

This investigation identified 169 significant signals of bleeding events tied to anticoagulant drugs (Supplemental Digital Content Table S3, available at: <http://links.lww.com/JS9/G314>). Multiple statistical measures, including ROR, PRR, EBGM, and BCPNN, were incorporated to evaluate the strength of these signals. The comprehensive approach facilitated a thorough assessment of the data, bolstering the credibility of the findings.

The most striking observation from the data was the consistently high signal strength for various types of hemorrhagic

Table 2**Top 30 significant signal strengths of AE for anticoagulant drugs ranked by ROR at the PT level**

PT	Cases	ROR (95%CI)	PRR (χ^2)	EBGM (EBGM05)	BCPNN (IC025)
Spinal subdural hemorrhage	5	144.79 (28.09–746.33)	144.79 (204)	42.08 (10.67)	5.4 (3.84)
Hemorrhagic arteriovenous malformation	629	126.58 (110.1–145.52)	126.49 (24 595.26)	40.41 (35.96)	5.34 (5.19)
Adrenal hemorrhage	238	59.18 (49.4–70.89)	59.16 (6731.86)	29.77 (25.59)	4.9 (4.67)
Mesenteric hemorrhage	37	57.92 (36.72–91.35)	57.92 (1034.79)	29.46 (20.12)	4.88 (4.31)
Basal ganglia hemorrhage	366	57.31 (49.6–66.22)	57.29 (10 176.57)	29.3 (25.96)	4.87 (4.69)
Bullous hemorrhagic dermatosis	59	56.96 (39.76–81.58)	56.95 (1635.22)	29.21 (21.63)	4.87 (4.41)
Lower gastrointestinal hemorrhage	3335	54.23 (51.72–56.85)	54.03 (89 815.03)	28.44 (27.33)	4.83 (4.77)
Pericardial hemorrhage	982	54.12 (49.61–59.04)	54.06 (26 453.19)	28.44 (26.45)	4.83 (4.72)
Retroperitoneal hemorrhage	1128	53.57 (49.4–58.09)	53.51 (30 211.56)	28.29 (26.44)	4.82 (4.72)
Diverticulum intestinal hemorrhagic	1001	51.91 (47.67–56.54)	51.86 (26 341.89)	27.83 (25.91)	4.8 (4.69)
Arteriovenous graft site hemorrhage	10	48.26 (20.85–111.71)	48.26 (252.46)	26.78 (13.27)	4.74 (3.69)
Chronic gastrointestinal bleeding	104	44.96 (34.8–58.08)	44.95 (2516.26)	25.75 (20.78)	4.69 (4.35)
Hemorrhagic hepatic cyst	36	44.36 (28.74–68.48)	44.36 (864.03)	25.55 (17.77)	4.68 (4.11)
Thalamus hemorrhage	349	44.34 (38.57–50.98)	44.33 (8372.36)	25.54 (22.73)	4.67 (4.49)
Thyroid hemorrhage	26	43.03 (25.9–71.47)	43.02 (612.35)	25.11 (16.42)	4.65 (3.99)
Gastrointestinal vascular malformation hemorrhagic	146	42.71 (34.49–52.9)	42.71 (3422.75)	25.01 (20.91)	4.64 (4.36)
Anastomotic ulcer hemorrhage	56	41.58 (29.5–58.62)	41.58 (1291.02)	24.62 (18.47)	4.62 (4.17)
Spinal cord hemorrhage	140	37.72 (30.49–46.67)	37.71 (3030.36)	23.23 (19.44)	4.54 (4.25)
Hemorrhage intracranial	5452	37.38 (36.13–38.68)	37.16 (116 894.77)	23.03 (22.38)	4.53 (4.48)
Gastrointestinal hemorrhage	29 505	36.44 (35.91–36.97)	35.28 (611 618.41)	22.3 (22.03)	4.48 (4.46)
Traumatic intracranial hemorrhage	309	36.02 (31.25–41.52)	36.01 (6485.32)	22.59 (20.06)	4.5 (4.3)
Internal hemorrhage	3919	35.51 (34.12–36.95)	35.36 (81 261.78)	22.33 (21.6)	4.48 (4.43)
Hemorrhagic cholecystitis	28	35.25 (22.04–56.4)	35.25 (579.29)	22.29 (15.05)	4.48 (3.85)
Epidural hemorrhage	43	32.77 (22.54–47.63)	32.77 (845.82)	21.29 (15.57)	4.41 (3.9)
Cerebellar hemorrhage	483	32.51 (29.08–36.34)	32.49 (9444.04)	21.17 (19.29)	4.4 (4.25)
Upper gastrointestinal hemorrhage	5520	32.47 (31.42–33.56)	32.28 (107 460.44)	21.08 (20.51)	4.4 (4.35)
Hemorrhagic stroke	2472	30.16 (28.73–31.66)	30.08 (45 744.05)	20.14 (19.34)	4.33 (4.26)
Hemorrhagic erosive gastritis	106	30.1 (23.8–38.06)	30.09 (1962.03)	20.15 (16.55)	4.33 (4.01)
Intraventricular hemorrhage	790	29.64 (27.2–32.29)	29.61 (14 452.93)	19.93 (18.55)	4.32 (4.2)
Subdural hemorrhage	842	29.07 (26.76–31.58)	29.04 (15 186.08)	19.68 (18.36)	4.3 (4.18)

AEs, adverse events; BCPNN, Bayesian Confidence Propagation Neural Network; EBGM, Empirical Bayes Geometric Mean; PRR, Proportional Reporting Ratio; PT, preferred term; ROR, Reporting Odds Ratio.

events (Table 2). Spinal subdural hemorrhage tops the list with an extremely high ROR of 144.79 (95% CI: 28.09–746.33), indicating a strong association between anticoagulant use and this serious AE. Other bleeding-related terms such as hemorrhagic arteriovenous malformation, adrenal hemorrhage, and mesenteric hemorrhage also exhibited ROR values well above 50, suggesting that these were significant safety concerns. Gastrointestinal bleeding appeared frequently in the list, with both upper and lower gastrointestinal hemorrhage showing substantial signal strength. Gastrointestinal hemorrhage in general had an ROR of 36.44 (95% CI: 35.91–36.97) with 29 505 cases.

The study also identified various hemorrhagic events associated with anticoagulant drugs, including bullous hemorrhagic dermatosis, anastomotic ulcer hemorrhage, spinal cord hemorrhage, intracranial hemorrhage (both traumatic and non-traumatic), internal hemorrhage, hemorrhagic cholecystitis, epidural hemorrhage, cerebellar hemorrhage, hemorrhagic stroke, hemorrhagic erosive gastritis, intraventricular hemorrhage, and subdural hemorrhage. Each of these events exhibited ROR values ranging from approximately 30 to 57, indicating a notable association with anticoagulant use. Some bleeding events, while having lower case counts, exhibited remarkably high signal strengths. For instance, hemorrhagic hepatic cyst (ROR 44.36, 95% CI: 28.74–68.48) and thalamus hemorrhage (ROR 44.34, 95% CI: 38.57–50.98), despite having relatively few reports, show ROR values in the mid-40s, indicating potential serious risks despite their rarity.

Drug-specific bleeding signal at the PT level of each anticoagulant drug

To translate spontaneous-reporting data into clinically actionable information, we further extracted the 10 highest-ranking bleeding PTs for each anticoagulant according to ROR (Table 3). All the signals were listed only when detected by all four algorithms concurrently (ROR, PRR, EBGM, and BCPNN). The aim was to identify which anatomical sites or hemorrhage patterns generate the strongest disproportionate signal for every drug, thereby guiding clinicians to focus surveillance on the most “drug-specific” bleeding locations.

In our analysis, the “Gastric hemorrhage,” “Lower Gastrointestinal hemorrhage,” or “Gastrointestinal hemorrhage” appears in the top-10 list of every OAC (warfarin, rivaroxaban, apixaban, edoxaban, and dabigatran) and even some parenteral drugs, confirming the gastrointestinal tract as the significant frequent real-world bleeding source regardless of the mechanism. Retroperitoneal bleeding was a shared high-signal event among heparins and parenteral direct thrombin inhibitors (argatroban and bivalirudin). Pericardial bleeding ranks specifically within the top decile for apixaban, dabigatran, fondaparinux, and bivalirudin. Betrixaban exhibited extreme RORs for traumatic (ROR 18 059.67, 95% CI: 8829.21–36 940.05) and intracranial (ROR 643.97, 95% CI: 249.82–1660) bleeding, although based on very small case numbers.

Table 3**Top 10 significant signal strengths of AEs for each anticoagulant drug ranked by ROR at the PT level**

Anticoagulant drugs	PT	Cases	ROR (95%CI)	PRR (χ^2)	EBGM (EBGM05)	BCPNN (IC025)
Warfarin	Arteriovenous graft site hemorrhage	6	171 (66.91–437)	170.99 (737.43)	124.63 (48.77)	6.96 (1.49)
	Thyroid hemorrhage	12	111.68 (59.4–209.96)	111.67 (1057.19)	89.9 (47.81)	6.49 (2.64)
	Subarachnoid hemorrhage neonatal	4	96 (32.66–282.18)	95.99 (310.62)	79.47 (27.04)	6.31 (0.83)
	Central nervous system hemorrhage	57	71.05 (53.74–93.92)	71.01 (3404.31)	61.58 (46.58)	5.94 (4.51)
	Hemorrhagic cholecystitis	9	65.14 (32.4–130.98)	65.14 (497.34)	57.12 (28.41)	5.84 (2.14)
	Bleeding time shortened	3	59.48 (17.86–198.09)	59.47 (152.57)	52.73 (15.83)	5.72 (0.37)
	Choroidal hemorrhage	20	57.01 (35.81–90.74)	57 (978.02)	50.77 (31.9)	5.67 (3.25)
	Gastric hemorrhage	1027	48.57 (45.53–51.81)	48.16 (42 902.9)	43.65 (40.92)	5.45 (5.29)
	Muscle hemorrhage	297	43.91 (38.97–49.46)	43.8 (11 332.91)	40.05 (35.55)	5.32 (4.97)
	Spinal cord hemorrhage	30	42.49 (29.23–61.78)	42.48 (1111.6)	38.95 (26.79)	5.28 (3.59)
Rivaroxaban	Hemorrhagic arteriovenous malformation	581	352.07 (307.84–402.66)	351.3 (74 502.81)	129.6 (113.31)	7.02 (6.58)
	Gastrointestinal vascular malformation hemorrhagic	206	126.15 (106.03–150.09)	126.05 (15 789.33)	78.26 (65.78)	6.29 (5.59)
	Anastomotic ulcer hemorrhage	48	115.09 (80.79–163.96)	115.07 (3468.82)	73.9 (51.87)	6.21 (4.41)
	Diverticulum intestinal hemorrhagic	749	113.88 (104.13–124.53)	113.56 (53 661.17)	73.28 (67.01)	6.2 (5.94)
	Chronic gastrointestinal bleeding	73	89.63 (68.06–118.04)	89.61 (4442.56)	62.54 (47.49)	5.97 (4.71)
	Basal ganglia hemorrhage	220	86.61 (73.97–101.42)	86.54 (13 056.64)	61.04 (52.13)	5.93 (5.36)
	Internal hemorrhage	2897	81.46 (78.02–85.06)	80.58 (163 186.99)	58.03 (55.57)	5.86 (5.77)
	Mesenteric hemorrhage	20	75.47 (45.18–126.07)	75.47 (1072.44)	55.34 (33.13)	5.79 (3.23)
	Hemorrhagic erosive gastritis	81	71.78 (55.72–92.47)	71.76 (4179.79)	53.33 (41.4)	5.74 (4.66)
	Lower gastrointestinal hemorrhage	1658	64.69 (61.21–68.37)	64.29 (78 536.79)	49.11 (46.47)	5.62 (5.5)
Apixaban	Hemorrhagic hepatic cyst	14	43.89 (24.67–78.07)	43.88 (485.29)	36.47 (20.5)	5.19 (2.62)
	Hemorrhagic cholecystitis	12	42 (22.6–78.07)	42 (400.27)	35.17 (18.92)	5.14 (2.4)
	Epidural hemorrhage	16	33.6 (19.82–56.96)	33.6 (436.3)	29.11 (17.17)	4.86 (2.7)
	Intestinal varices hemorrhage	3	31.5 (9.36–106.02)	31.5 (77.04)	27.52 (8.18)	4.78 (0.29)
	Spinal cord hemorrhage	40	26.93 (19.38–37.43)	26.92 (885.02)	23.98 (17.25)	4.58 (3.46)
	Traumatic intracranial hemorrhage	88	24.42 (19.58–30.46)	24.41 (1770.26)	21.98 (17.62)	4.46 (3.83)
	Anal ulcer hemorrhage	3	24.23 (7.33–80.06)	24.23 (59.91)	21.83 (6.61)	4.45 (0.27)
	Thalamus hemorrhage	80	23.28 (18.48–29.33)	23.27 (1535)	21.05 (16.71)	4.4 (3.74)
	Pericardial hemorrhage	178	20.25 (17.36–23.62)	20.24 (2969.4)	18.55 (15.9)	4.21 (3.85)
	Spinal subarachnoid hemorrhage	5	20.19 (8.07–50.56)	20.19 (83.21)	18.51 (7.39)	4.21 ^[1]
Edoxaban	Putamen hemorrhage	9	134.26 (69.04–261.08)	134.17 (1148.78)	129.6 (66.65)	7.02 (2.29)
	Thalamus hemorrhage	21	101.57 (65.82–156.72)	101.42 (2033.35)	98.79 (64.02)	6.63 (3.56)
	Hemobilia	9	96.22 (49.64–186.51)	96.17 (826.51)	93.8 (48.39)	6.55 (2.26)
	Urogenital hemorrhage	7	77.69 (36.75–164.24)	77.65 (518.99)	76.11 ^[36]	6.25 (1.84)
	Hemorrhagic cerebral infarction	10	57.54 (30.81–107.46)	57.5 (546.82)	56.65 (30.33)	5.82 (2.35)
	Rectal ulcer hemorrhage	3	51.92 (16.61–162.25)	51.91 (147.74)	51.21 (16.39)	5.68 (0.46)
	Cerebral microhemorrhage	3	48.37 (15.48–151.07)	48.36 (137.37)	47.76 (15.29)	5.58 (0.46)
	Basal ganglia hemorrhage	9	46.59 (24.14–89.93)	46.57 (396.39)	46.01 (23.84)	5.52 (2.14)
	Hemorrhage subcutaneous	39	45.92 (33.47–63)	45.8 (1688.54)	45.26 (32.99)	5.5 (3.97)
	Cerebellar hemorrhage	16	45.53 (27.8–74.55)	45.48 (687.72)	44.95 (27.45)	5.49 (2.94)
Betrixaban	Traumatic hemorrhage	11	18 059.67 (8829.21–36 940.05)	12 384.09 (135 122.31)	12 285.52 (6006.29)	13.58 (2.62)
	Hemorrhage intracranial	5	643.97 (249.82–1660)	552.12 (2750.34)	551.92 (214.11)	9.11 (1.3)
	Gastrointestinal hemorrhage	11	317.22 (155.38–647.61)	217.84 (2377.34)	217.81 (106.69)	7.77 (2.55)
	Hemorrhage	7	150.9 (65.92–345.48)	120.92 (833.84)	120.91 (52.82)	6.92 (1.79)
Dabigatran	Lower gastrointestinal hemorrhage	1161	69.57 (65.31–74.12)	69.07 (64 810.24)	57.64 (54.1)	5.85 (5.69)
	Pericardial hemorrhage	267	52.06 (45.76–59.21)	51.97 (11 587.66)	45.25 (39.78)	5.5 (5.09)
	Mesenteric hemorrhage	9	47.38 (23.6–95.14)	47.38 (358.91)	41.74 (20.79)	5.38 (2.07)
	Gastrointestinal hemorrhage	7759	39.73 (38.79–40.7)	37.82 (250 836.5)	34.16 (33.35)	5.09 (5.05)
	Traumatic intracranial hemorrhage	85	38.29 (30.6–47.91)	38.27 (2774.91)	34.52 (27.59)	5.11 (4.31)
	Hemorrhagic transformation stroke	120	32.82 (27.21–39.58)	32.8 (3375.84)	30.02 (24.89)	4.91 (4.32)
	Hemorrhage intracranial	1196	32.24 (30.38–34.22)	32.01 (32 859.47)	29.35 (27.66)	4.88 (4.75)
	Diverticulitis intestinal hemorrhagic	31	31.96 (22.12–46.18)	31.95 (850.08)	29.31 (20.28)	4.87 (3.42)
	Thoracic hemorrhage	15	31.11 (18.34–52.78)	31.11 (400.67)	28.6 (16.86)	4.84 (2.64)
	Upper gastrointestinal hemorrhage	1241	30.24 (28.53–32.05)	30.01 (32 003.11)	27.67 (26.1)	4.79 (4.67)
Heparin	Bullous hemorrhagic dermatosis	17	258.79 (154.73–432.81)	258.66 (3729.34)	221.22 (132.27)	7.79 (3.33)
	Adrenal hemorrhage	35	122.26 (86.63–172.55)	122.14 (3892.8)	113.14 (80.17)	6.82 (4.28)
	Vascular access site hemorrhage	8	120.55 (58.68–247.64)	120.52 (878.62)	111.75 (54.39)	6.8 (2.07)

(Continues)

Table 3
(Continued).

Anticoagulant drugs	PT	Cases	ROR (95%CI)	PRR (χ^2)	EBGM (EBGM05)	BCPNN (IC025)
Enoxaparin	Mediastinal hemorrhage	6	102.59 (44.88–234.52)	102.58 (565.4)	96.16 (42.07)	6.59 (1.59)
	Hemorrhagic cholecystitis	4	89.51 (32.65–245.39)	89.5 (330.61)	84.59 (30.85)	6.4 (0.92)
	Graft hemorrhage	4	82.25 (30.07–224.98)	82.25 (304.57)	78.08 (28.55)	6.29 (0.92)
	Pulmonary hemorrhage neonatal	4	81.16 (29.68–221.9)	81.15 (300.61)	77.09 (28.19)	6.27 (0.92)
	Ureteric hemorrhage	3	78.71 (24.66–251.21)	78.7 (218.82)	74.88 (23.46)	6.23 (0.45)
	Retroperitoneal hemorrhage	110	75.15 (62.04–91.03)	74.92 (7646.43)	71.45 (58.99)	6.16 (5.17)
	Pancreatic hemorrhage	5	64.48 (26.35–157.8)	64.47 (299.74)	61.89 (25.29)	5.95 (1.27)
	Retroperitoneal hemorrhage	236	186.94 (163.33–213.95)	185.59 (38 967.56)	167 (145.92)	7.38 (6.42)
	Bullous hemorrhagic dermatosis	7	105.51 (49.14–226.54)	105.49 (681.16)	99.24 (46.22)	6.63 (1.85)
	Spinal epidural hemorrhage	3	77.71 (24.41–247.37)	77.71 (216.98)	74.27 (23.33)	6.21 (0.46)
	Muscle hemorrhage	136	69.59 (58.6–82.64)	69.31 (8788.61)	66.56 (56.05)	6.06 (5.24)
	Pelvic hemorrhage	6	58.18 (25.77–131.32)	58.17 (325.67)	56.23 (24.91)	5.81 (1.55)
	Abdominal wall hemorrhage	7	50.91 (23.99–108)	50.89 (332.2)	49.41 (23.29)	5.63 (1.77)
	Shock hemorrhagic	186	45.8 (39.58–53)	45.54 (7887.68)	44.35 (38.33)	5.47 (4.96)
	Spinal cord hemorrhage	8	38.56 (19.13–77.73)	38.55 (285.97)	37.7 (18.7)	5.24 (1.92)
Dalteparin	Intra-abdominal hemorrhage	49	38.1 (28.7–50.59)	38.05 (1727.96)	37.21 (28.03)	5.22 (4.02)
	Thoracic hemorrhage	4	37.68 (13.99–101.52)	37.68 (139.63)	36.86 (13.68)	5.2 (0.86)
	Abdominal wall hemorrhage	7	202.93 (95.63–430.64)	202.76 (1363.47)	196.75 (92.71)	7.62 (1.91)
	Adrenal hemorrhage	3	42.35 (13.61–131.81)	42.33 (120.3)	42.07 (13.52)	5.39 (0.45)
	Muscle hemorrhage	21	41.28 (26.86–63.43)	41.18 (818.18)	40.93 (26.64)	5.36 (3.24)
	Retroperitoneal hemorrhage	12	34.03 (19.29–60.04)	33.98 (382.2)	33.81 (19.17)	5.08 (2.46)
	Hemoperitoneum	8	27.08 (13.52–54.24)	27.05 (199.9)	26.95 (13.45)	4.75 (1.83)
	Procedural hemorrhage	14	25.63 (15.16–43.34)	25.59 (329.55)	25.5 (15.08)	4.67 (2.53)
	Intra-abdominal hemorrhage	8	24.3 (12.13–48.67)	24.28 (177.92)	24.19 (12.08)	4.6 (1.79)
	Hemorrhagic transformation stroke	5	24.17 (10.04–58.18)	24.16 (110.58)	24.07 ^[10]	4.59 (1.13)
	Shock hemorrhagic	25	23.89 (16.12–35.4)	23.82 (544.67)	23.74 (16.02)	4.57 (3.09)
	Arterial hemorrhage	3	20.29 (6.53–63.02)	20.28 (54.82)	20.22 (6.51)	4.34 (0.35)
	Bullous hemorrhagic dermatosis	6	654.39 (287.63–1488.82)	653.52 (3708.65)	620.05 (272.54)	9.28 (1.67)
	Muscle hemorrhage	26	94.01 (63.84–138.44)	93.47 (2360.41)	92.76 (62.99)	6.54 (3.84)
	Shock hemorrhagic	52	92.12 (70.01–121.21)	91.06 (4597.69)	90.39 (68.7)	6.5 (4.67)
Fondaparinux	Pancreatitis hemorrhagic	3	83.05 (26.67–258.62)	83 (241.38)	82.44 (26.47)	6.37 (0.5)
	Retroperitoneal hemorrhage	15	78.13 (46.98–129.91)	77.87 (1130.95)	77.37 (46.53)	6.27 (3.02)
	Arterial hemorrhage	4	49.59 (18.57–132.46)	49.55 (189.51)	49.35 (18.48)	5.63 (0.91)
	Hemorrhagic disorder	3	36.66 (11.8–113.91)	36.64 (103.68)	36.53 (11.76)	5.19 (0.44)
	Pericardial hemorrhage	5	29.96 (12.45–72.09)	29.93 (139.46)	29.85 (12.41)	4.9 (1.18)
	Small Intestinal hemorrhage	4	26.89 (10.08–71.76)	26.87 (99.4)	26.81 (10.05)	4.74 (0.83)
	Hemoperitoneum	4	24.73 (9.27–65.99)	24.71 (90.83)	24.66 (9.24)	4.62 (0.81)
	Thoracic hemorrhage	5	340.93 (140.08–829.74)	340.55 (1645.76)	331.12 (136.05)	8.37 (1.37)
	Puncture site hemorrhage	17	331.25 (204.4–536.82)	330.01 (5426.11)	321.15 (198.17)	8.33 (3.4)
	Anastomotic hemorrhage	3	209.25 (66.8–655.49)	209.11 (610.64)	205.52 (65.61)	7.68 (0.52)
	Retroperitoneal hemorrhage	30	155.55 (108.38–223.26)	154.53 (4517.68)	152.56 (106.3)	7.25 (4.17)
	Medical device site hemorrhage	3	153.57 (49.15–479.81)	153.47 (448.65)	151.53 (48.5)	7.24 (0.52)
	Respiratory tract hemorrhage	5	105.78 (43.84–255.26)	105.67 (513.84)	104.75 (43.41)	6.71 (1.33)
	Catheter site hemorrhage	18	101.84 (63.98–162.11)	101.44 (1775.08)	100.59 (63.2)	6.65 (3.34)
	Pharyngeal hemorrhage	7	76.66 (36.44–161.29)	76.55 (518.59)	76.06 (36.16)	6.25 (1.85)
Bivalirudin	Procedural hemorrhage	16	53 (32.4–86.68)	52.81 (809.78)	52.58 (32.15)	5.72 ^[3]
	Pericardial hemorrhage	8	47.36 (23.64–94.89)	47.28 (360.95)	47.09 (23.5)	5.56 (1.98)
	Hemorrhagic cerebral infarction	12	466.22 (262.99–826.51)	463.61 (5439.64)	455.28 (256.82)	8.83 (2.85)
	Bronchial hemorrhage	4	340.93 (127–915.22)	340.3 (1335.22)	335.79 (125.08)	8.39 ^[1]
	Hemorrhagic infarction	6	302.84 (135.25–678.08)	301.99 (1778.75)	298.44 (133.29)	8.22 (1.68)
	Bleeding time prolonged	6	134.54 (60.24–300.44)	134.16 (788.84)	133.46 (59.76)	7.06 (1.65)
	Catheter site hemorrhage	5	59.5 (24.72–143.26)	59.37 (286.26)	59.23 (24.6)	5.89 (1.29)
	Subarachnoid hemorrhage	17	47.04 (29.17–75.85)	46.67 (758.54)	46.59 (28.89)	5.54 (3.04)
	Pulmonary alveolar hemorrhage	8	43.54 (21.73–87.22)	43.38 (330.66)	43.3 (21.62)	5.44 (1.96)
	Intraventricular hemorrhage	3	32.58 (10.49–101.16)	32.53 (91.58)	32.49 (10.46)	5.02 (0.43)
	Procedural hemorrhage	4	27.95 (10.47–74.57)	27.9 (103.62)	27.87 (10.44)	4.8 (0.84)
	Pulmonary hemorrhage	6	21.13 (9.48–47.1)	21.07 (114.64)	21.06 (9.45)	4.4 (1.35)
Argatroban						

AEs, adverse events; BCPNN, Bayesian Confidence Propagation Neural Network; EBGM, Empirical Bayes Geometric Mean; PRR, Proportional Reporting Ratio; PT, preferred term; ROR, Reporting Odds Ratio.

There were also several drug-specific bleeding events. Warfarin had the strongest signals cluster in “surgical” or diffuse sites (arteriovenous graft site, thyroid, cholecystitis, and neonatal subarachnoid hemorrhage), which were consistent with its long half-life, widespread use in vascular-access procedures, and need for bridging therapy. Rivaroxaban had the highest RORs involving structural vascular malformations (arteriovenous malformation and diverticular bleeding), solid-organ mesenteric, or basal-ganglia hemorrhage, aligning with its high peak concentration and extensive gut-wall penetration. The signals of apixaban are generally lower; the top events are hepatic cyst, cholecystitis, and intraspinal/epidural sites, suggesting a comparatively favorable overall profile but caution around hepatobiliary and neuraxial procedures. Edoxaban had thalamic, putaminal, and urogenital bleeding that dominated, which was possibly linked to its renal-dependent clearance and higher plasma levels. Dabigatran had lower gastrointestinal, diverticular, and thoracic hemorrhage predominance, matching its propensity for gastro-duodenal erosions and the observation that chest wall trauma while on dabigatran produces large hematomas. Heparin had unique prominence of access-site, adrenal, mediastinal, and neonatal pulmonary hemorrhage, which occurred at sites exposed to high heparin concentrations during cardiac surgery or catheter insertion. Parenteral direct thrombin inhibitors (argatroban and bivalirudin) had extremely high RORs for intracranial, spinal epidural, and pulmonary alveolar bleeding, underscoring the need for meticulous neuraxial and airway monitoring during surgery therapy.

As for the organ-system distribution differences, warfarin, rivaroxaban, and argatroban had the strongest central nervous system signal. Hepatobiliary bleeding was uniquely prominent for apixaban and warfarin. Retroperitoneum and adrenal bleeding were dominant signal classes for all heparins and fondaparinux. Lower-gastrointestinal bleeding signal was most prominent in dabigatran and rivaroxaban. Bullous hemorrhagic dermatosis and muscle bleeding were specific top signals for heparin and fondaparinux, rarely seen with oral agents.

Difference of bleeding risk signals under subgroup populations

Subgroup analysis of age. Subgroup analyses were integral to signal refinement in disproportionality analyses, allowing detection of effect-measure modification that may be obscured in aggregate data. We therefore performed five *a priori* stratifications by age (>65 vs. ≤65 years), sex, reporter qualification, and comorbidities of hypertensive and gastric ulcer, to test the robustness of bleeding signals associated with anticoagulants and to identify sub-populations at disproportionate risk (Fig. 5A). Age exhibited the strongest heterogeneity. Among elderly patients (>65 years), the most prominent event was heavy menstrual bleeding (ROR 57.52; 95% CI 49.05–67.45), whereas subjects ≤65 years displayed markedly higher reporting for intracranial and visceral hemorrhagic events, exemplified by hemorrhagic arteriovenous malformation (ROR 240.13; 175.1–329.32), anastomotic ulcer hemorrhage (ROR 111.64; 70.92–175.73), and diverticular intestinal hemorrhage (ROR 108.29; 87.83–133.51). The inversion of the menstrual-bleeding signal in older cohorts, together with the predominance of life-threatening anatomical sites in younger adults, indicates age-dependent susceptibility patterns that transcend mere prevalence

differences and suggest distinct pharmacodynamic or comedication mechanisms meriting prospective validation.

Subgroup analysis of sex. Sex-stratified disproportionality mining revealed divergent bleeding signatures linked to anticoagulants (Fig. 5B). In women, high-order signals clustered around rare but catastrophic vascular lesions, with hemorrhagic arteriovenous malformation generating the largest reporting excess (ROR 206.1; 95% CI 164.11–258.83), followed by arteriovenous graft-site hemorrhage (ROR 108.4; 95% CI 30.59–384.15) and anastomotic ulcer bleeding (ROR 91.01; 95% CI 54.91–150.84). In contrast, men exhibited a predilection for deep solid-organ and intracranial territories such as mesenteric (ROR 62.01; 95% CI 31.96–120.28), basal ganglia (ROR 52.68; 95% CI 43.21–64.23) and thalamic hemorrhage (ROR 39.22; 95% CI 32.54–47.27), while maintaining comparatively attenuated overall RORs. Lower gastrointestinal hemorrhage, the commonest event in both sexes, remained more strongly associated with female reports (ROR 77.75 vs. ROR 40.95). These opposing gradients, rare arteriovenous ruptures in women versus parenchymal microbleeding in men, implicate sex-specific hemostatic or anatomical vulnerabilities that transcend traditional bleeding-risk covariates and warrant mechanistic interrogation.

Subgroup analysis of reporters. Reporter-stratified disproportionality analysis revealed pronounced differences in bleeding signals between medical personnel reporters and nonmedical personnel (Fig. 5C). Among healthcare professionals (physicians, pharmacists, nurses, and other clinicians), spinal subdural hemorrhage generated the highest reporting excess (ROR 194.56; 95% CI 21.75–1740.78), followed by well-localized, anatomically discrete lesions within the basal ganglia (ROR 48.87; 95% CI 40.9–58.39), pericardium (ROR 41.55; 95% CI 37.25–46.34), and spinal cord (ROR 37.84; 95% CI 29.06–49.27). In contrast, nonmedical personnel (consumer or legal reporters) produced substantially larger effect sizes for life-threatening visceral and iatrogenic events, including mesenteric hemorrhage (ROR 195.42; 95% CI 71.59–533.44), hemorrhagic arteriovenous malformation (ROR 161.92; 95% CI 137.55–190.61), lower gastrointestinal bleeding (ROR 130.3; 95% CI 120.48–140.93), and anastomotic ulcer hemorrhage (ROR 111.52; 95% CI 67.07–185.43). The narrower confidence intervals and lower point estimates observed for medically reported events imply earlier recognition and more complete documentation of sentinel cases, whereas the exaggerated RORs among nonmedical reporters reflect selective submission of dramatic, often litigation-associated outcomes. These complementary but divergent signal landscapes underscore the necessity of triangulating data sources and reporter perspectives when constructing pharmacovigilance alerts for anticoagulant-related bleeding.

Subgroup analysis of comorbidities of hypertension and gastric ulcer. Hypertension-comorbidity stratification unveiled a steep dose-response-like amplification of anticoagulant-related bleeding signals (Fig. 5D). In patients with documented hypertension, hemorrhagic arteriovenous malformation dominated the safety landscape (ROR 127.79; 95% CI 111.01–147.11), followed by spinal subdural (ROR 76.55; 95% CI 17.13–342.03), basal ganglia (ROR 60.33; 95% CI 51.97–70.04), pericardial (ROR 57.54; 95% CI 52.56–62.99), and lower gastrointestinal hemorrhage (ROR 56.37; 95% CI 53.7–59.16), all

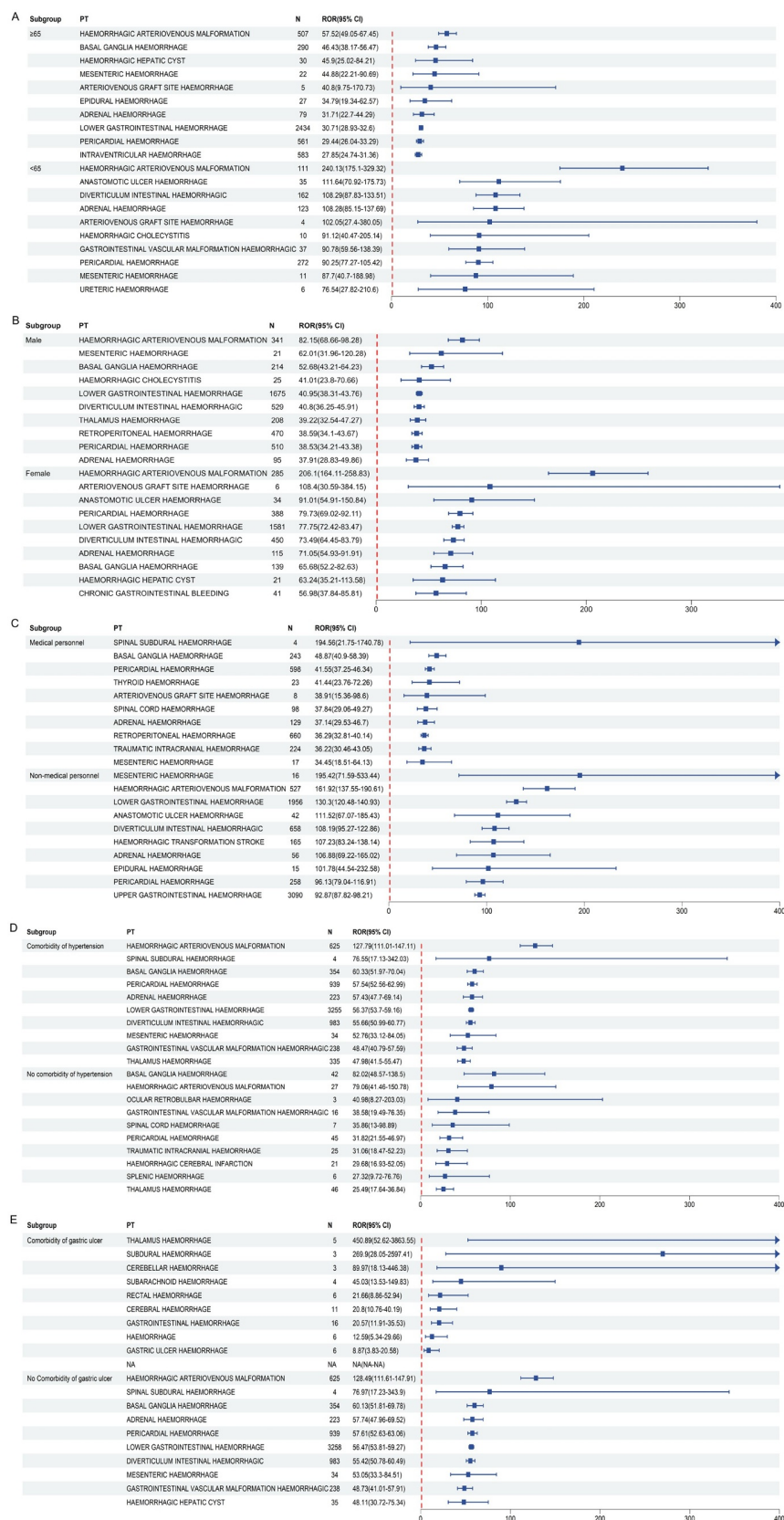


Figure 5. Top 10 significant signal strengths of AEs for anticoagulant drugs ranked by ROR at the PT level under different subgroup populations. AEs, adverse events; PT, preferred term; ROR, Reporting Odds Ratio.

clustering tightly within a 50- to 130-fold excess risk bandwidth. Conversely, non-hypertensive comparables generated meaningfully lower and more dispersed RORs (peak 82.02; 95% CI 48.57–138.5 for basal ganglia), with no single PT exceeding an 80-fold threshold. The uniformity of elevated point estimates across disparate anatomical territories in the hypertensive cohort implies a global, pressure-mediated disruption of vascular integrity that synergizes with anticoagulant exposure, rather than a focal lesion-specific effect. These data support the hypothesis that hypertension functions as an independent, multisite effect modifier of bleeding propensity and underscores the need for precision-dosing algorithms that integrate ambulatory blood-pressure trajectories beyond crude comorbidity flags.

Comorbidity-stratified pharmacovigilance disclosed a profound anatomical shift in anticoagulant-related bleeding when pre-existing gastric ulcer was present (Fig. 5E). Among ulcer-positive subjects, disproportionality was concentrated in the posterior fossa: thalamic hemorrhage dominated with an ROR of 450.89 (95% CI 52.62–3863.55), followed by cerebellar (ROR 89.97; 18.13–446.38) and subarachnoid (ROR 45.03; 95% CI 13.53–149.83) events, which were territories seldom highlighted in aggregate analyses. Conversely, the non-gastric ulcer cohort reproduced the conventional “deep-visceral” pattern, led by hemorrhagic arteriovenous malformation (ROR 128.49; 95% CI 111.61–147.91), spinal subdural (ROR 76.97; 95% CI 17.23–343.9), and basal ganglia bleeding (ROR 60.13; 95% CI 51.81–69.78). It implies that compromised gastric mucosal integrity may act as a sentinel marker for occult cerebral micro-angiopathy. These data mandate routine neuro-imaging surveillance in anticoagulated patients with gastric ulcer disease and encourage investigation of gastro-cerebral vascular crosstalk beyond traditional bleeding risk scores.

Logistic regression analyses of bleeding signal associated with anticoagulant drugs

Multivariable logistic regression across four hierarchical models identified agents with distinct, model-independent bleeding risk

profiles relative to warfarin (Table 4). Rivaroxaban consistently exhibited a 47–51% higher odds of reporting (OR 1.47–1.51; 95% CI nonoverlapping 1.00), irrespective of incremental covariate adjustment, corroborating a genuine class effect of direct factor-Xa inhibition. Conversely, apixaban seemed the safest OAC, maintaining a 67–69% risk reduction (OR 0.31–0.33) across all specifications, while edoxaban produced a stable, intermediate 40% decrease (OR 0.59–0.61). Among parenteral agents, low-molecular-weight heparins (enoxaparin and dalteparin) and fondaparinux yielded concordant 30–55% risk attenuations (OR 0.45–0.75), whereas the oral direct thrombin inhibitor dabigatran converged to the null effect (OR 1.02–1.07) after confounder control. Notably, betrixaban displayed extreme, albeit imprecise, excess odds (OR 71.8–92.6) reflective of sparse-event inflation. Stability of estimates from crude to fully adjusted models (age, sex, reporter qualification, hypertension, and gastric ulcer) indicates robustness against selection and indication bias, providing high-granularity, real-world benchmarks for regulatory and clinical benefit-risk evaluations of contemporary anticoagulants.

Discussion

Thrombosis is widely recognized as a common pathological mechanism underlying various thrombotic diseases, including coronary syndrome, ischemic stroke, and peripheral vascular disease^[41]. As the pace of population aging intensifies and lifestyles continue to evolve, thromboembolic diseases have emerged as a more significant global health challenge^[42]. The increasingly extensive anticoagulant therapy and the increasing number of anticoagulant drugs make DRP increasingly prominent^[43]. Lack of awareness of anticoagulant therapy and effective patient management, as well as poor adherence, may break the balance between thrombus and bleeding, leading to serious AEs such as bleeding complications or thrombosis recurrence^[44]. In this study, we not only conducted a narrative review summarizing the evidence on bleeding complications related to anticoagulant drugs

Table 4
Logistic regression analyses of bleeding signal associated with anticoagulant drugs

Anticoagulant drugs	OR (95% CI)			
	Model 1	Model 2	Model 3	Model 4
Warfarin	Reference	Reference	Reference	Reference
Rivaroxaban	1.47 (1.43, 1.50)*	1.51 (1.47, 1.54)*	1.50 (1.46, 1.53)*	1.47 (1.43, 1.50)*
Apixaban	0.33 (0.32, 0.34)*	0.32 (0.32, 0.33)*	0.32 (0.31, 0.33)*	0.31 (0.31, 0.32)*
Edoxaban	0.57 (0.53, 0.61)*	0.60 (0.56, 0.64)*	0.61 (0.57, 0.65)*	0.59 (0.56, 0.64)*
Betrixaban	71.8 (15.5, 1277)*	92.0 (19.9, 1636)*	92.6 (12.7, 676)*	90.3 (12.4, 660)*
Dabigatran	1.07 (1.04, 1.09)*	1.03 (1.00, 1.05)	1.03 (1.00, 1.06)	1.02 (0.99, 1.04)
Heparin	0.46 (0.44, 0.49)*	0.51 (0.48, 0.53)*	0.51 (0.48, 0.54)*	0.51 (0.48, 0.53)*
Dalteparin	0.45 (0.41, 0.50)*	0.48 (0.43, 0.53)*	0.48 (0.44, 0.54)*	0.49 (0.44, 0.54)*
Enoxaparin	0.60 (0.57, 0.63)*	0.66 (0.63, 0.69)*	0.66 (0.63, 0.69)*	0.67 (0.64, 0.70)*
Fondaparinux	0.68 (0.61, 0.77)*	0.75 (0.66, 0.84)*	0.74 (0.66, 0.84)*	0.73 (0.65, 0.82)*
Argatroban	0.70 (0.59, 0.81)*	0.74 (0.63, 0.86)*	0.75 (0.64, 0.88)*	0.74 (0.64, 0.87)*
Bivalirudin	0.77 (0.69, 0.87)*	0.86 (0.77, 0.97)*	0.87 (0.78, 0.98)*	0.87 (0.77, 0.97)*

CI, confidence interval; OR, odds ratio. The asterisk indicates statistical significance.
Model 1: crude ROR.
Model 2: adjusted for age and sex.
Model 3: adjusted for age, sex, and reporter.
Model 4: adjusted for age, sex, reporter, and comorbidities (hypertension and gastric ulcer).
*P < 0.001, ** P < 0.05, significant difference compared with warfarin.

but also analyzed the bleeding event reports of FAERS, which facilitate anticoagulant treatment management, drawing on clinical insights and real-world data.

Comparing real-world data with previous studies

Our research revealed significant variations in bleeding risks of anticoagulants, with elderly patients over 65 years old being particularly susceptible to bleeding complications. Our results were consistent with a previous study, that is, elderly patients are particularly vulnerable to bleeding events related to anticoagulant therapy^[45]. It was reported that in the HAS-BLED framework, when advanced age (>65 years) is combined with other risk factors [hypertension, abnormal renal/liver function, stroke, bleeding history or predisposition, labile international normalized ratio (INR), elderly, and drugs/alcohol concomitantly], patients would be particularly prone to NOAC-related bleeding, regardless of the specific anticoagulant agent used^[46]. Besides, our data showed that male patients accounted for 47.3% of the bleeding-related AE reports, while female patients made up 41.9%. This suggests that both genders are similarly affected by anticoagulant-related bleeding events. Nearly half (46.8%) of the cases required hospitalization, and 20.7% resulted in death. It is a bit higher than the risk of death after NOAC-related bleeding of 17.7% (95% CI: 15.1–20.4%)^[47]. This underscores the serious nature of bleeding events associated with anticoagulant therapy and the need for careful management and monitoring.

Our data revealed a wide range of bleeding types, including intracranial hemorrhage and gastrointestinal bleeding, which were also reported in a multinational population-based cohort study^[48]. This diversity highlights the complexity of managing anticoagulant therapy and the need for tailored approaches based on patient-specific factors. Several bleeding events were with high signal strength (Table 2), such as spinal subdural hemorrhage (ROR 144.79) and gastrointestinal hemorrhage (ROR 36.44). These high ROR values indicate a strong association between anticoagulant use and these severe bleeding events. However, spinal subdural hemorrhage is a rare medical emergency that can cause permanent neurological deficits^[49]. A recent study showed a case of spontaneous subdural spinal hemorrhages associated with rivaroxaban use^[49]. This alerts clinicians to consider acute neurological deficit as a possible spinal bleed, justifies early MRI imaging, and underlines the need for prompt reversal and surgical decompression. For anticoagulant-associated gastrointestinal hemorrhage, moderate-certainty evidence showed a probable association with older age, kidney disease, concomitant use of aspirin, concomitant use of the antiplatelet agent, heart failure, myocardial infarction, hematochezia, renal failure, coronary artery disease, *Helicobacter pylori* infection, social risk factors, alcohol use, smoking, anemia, history of sleep apnea, chronic obstructive pulmonary disease, INR, and obesity^[50].

Real-world data analysis poses challenges to the bleeding risks management

This study further integrated the disproportionality signals across five clinically relevant strata (age, sex, reporter qualification, hypertension, and gastric-ulcer status) with the fully adjusted logistic estimates and yields a coherent hierarchy of anticoagulant-related bleeding risk that is both internally consistent and clinically interpretable (Table 4 and Fig. 5). The

subgroup dissections revealed effect-measure modification of similar magnitude to the between-drug differences detected in the regression models: for example, the three- to four-fold ROR amplification observed for hemorrhagic arteriovenous malformation in hypertensive or female patients parallels the 3-fold excess odds conferred by rivaroxaban relative to apixaban (OR 1.47 vs. 0.31), indicating that patient-level and drug-level determinants contribute independently and multiplicatively to the bleeding phenotype. Conversely, the extreme but sparse posterior-fossa signals generated in the gastric-ulcer stratum (thalamic ROR 451) were not driven by any single anticoagulant, yet were captured only when anatomical specificity was preserved – an insight unobtainable from aggregate logistic output. Stability of agent-specific ORs across Models 1–4 ($\leq 5\%$ change for apixaban, rivaroxaban, and edoxaban) argues against unmeasured confounding by the same covariates that dramatically reshaped ROR rankings in the sensitivity analyses, reinforcing the validity of both analytical layers. Collectively, these complementary approaches demonstrate that robust pharmacovigilance requires simultaneous examination of “who bleeds” and “with which drug”: subgroup disproportionality supplies anatomical precision and identifies susceptible phenotypes, while multivariable logistic regression quantifies comparative safety and preserves statistical power. Triangulating the two frameworks positions apixaban as the safest oral agent across all phenotypes, rivaroxaban as consistently pro-hemorrhagic, and betrixaban as an outlier requiring enhanced post-marketing surveillance, thereby offering actionable, high-resolution evidence for regulators, trialists, and clinicians.

Emerging role of multidisciplinary medical teams in the anticoagulation management process

As we know, nurses serve as essential caregivers, orchestrating a multifaceted array of responsibilities that encompass patient symptom monitoring, health education, inpatient drug administration, and rehabilitation support^[51–54]. It was reported that their vigilant oversight of patients' vital signs and coagulation profiles is pivotal in preemptively identifying potential bleeding risks and adverse reactions^[51,55]; the routine assessment of the INR in patients undergoing warfarin therapy exemplifies their role in ensuring precise dosing and optimizing therapeutic efficacy^[56]. They meticulously evaluate patients' clinical status, thereby providing physicians with real-time, data-driven insights that inform timely and context-specific treatment adjustments^[53]. The collaborative approach ensures that therapeutic strategies are finely tuned to individual patient needs. Nurses, who serve as primary educators for patients and their families, were reported to contribute to a significant reduction in the incidence of adverse reactions, thereby enhancing the overall efficacy and safety of anticoagulant therapy^[57]. Nurses also provide detailed instruction on optimal subcutaneous injection techniques to reduce the risk of local bleeding and promote enhanced procedural safety^[58]. In the context of PCI, nurses play a pivotal role in enhancing the quality of anticoagulation management and bolstering procedural safety through the implementation of rigorous, comprehensive quality control measures^[59]. Nurses facilitate patient recovery through a multifaceted approach that includes meticulous wound care, evidence-based dietary guidance, and targeted rehabilitation training. In patients at high risk of bleeding, nurses proactively implement timely interventions to mitigate complications, thereby

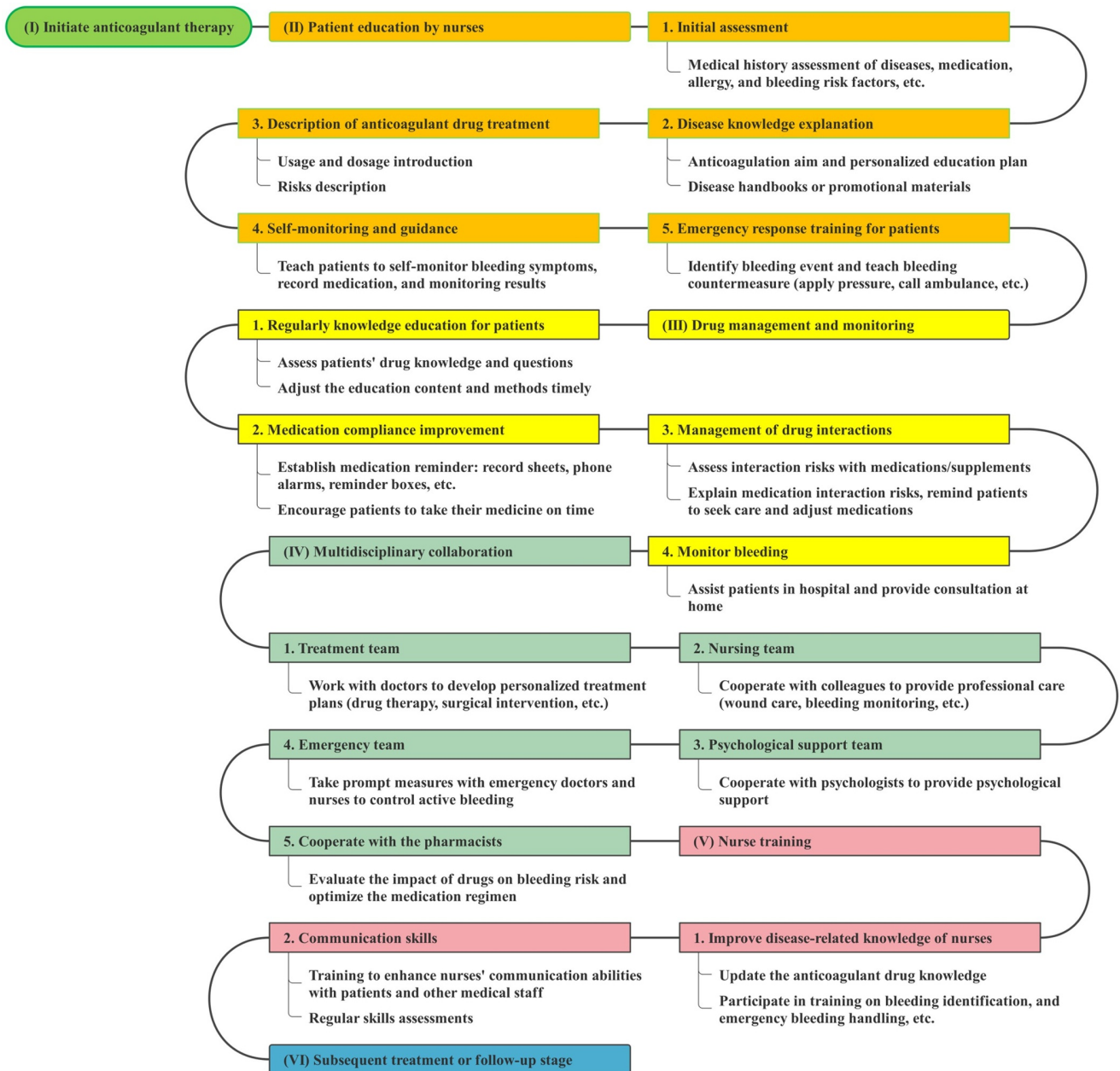


Figure 6. Role of nurses involvement in anticoagulant management.

preventing exacerbation of conditions and providing critical support to physicians in the delivery of effective treatment^[58].

Anticoagulation therapy also showed some efficacy in patients with repeated implantation failure and hypercoagulability^[60]. Low-molecular-weight heparin is extensively used in assisted reproduction and pregnancy field, with anticoagulant, anti-inflammatory, and immunomodulatory effects^[61]. Clinically, it is used to prevent and treat thromboembolic events, improve embryo implantation, and reduce placenta-related complications like preeclampsia and fetal growth restriction^[61]. Nurses are helpful in managing bleeding risks during anticoagulation therapy, closely monitoring the coagulation status and identifying bleeding risk factors including patient age, weight, comorbidities, and

potential drug interactions, and then collaborate with physicians to adjust treatment regimens if needed^[62]. In a published study from a reproductive center, nurses significantly contributed to enhancing clinical pregnancy rates among patients in frozen embryo transfer cycles by administering low-molecular-weight heparin calcium for anticoagulation, alongside providing psychological support, personalized health education, and effective nursing interventions, thereby achieving higher pregnancy rates compared to the control group^[63].

In the context of anticoagulant therapy, the role of nursing care is indispensable^[64], from patient education, identifying potential ADR risks, to necessary precautions (Fig. 6)^[51,53,54]. Besides, in the realm of anticoagulation therapy, doctors, pharmacists, and

Table 5
Emerging specific antagonists of anticoagulant drugs

Anticoagulant drug	Specific reversal agents	Mechanisms of reversal agents	FDA approve time
Heparin	Protamine	Dissociating the heparin-antithrombin III complex and forming the inactive heparin-protamine complex	1983
Warfarin	Vitamin K1	Inhibiting vitamin K epoxide reductase and interfering with the α carboxylation process of coagulation factors II, VII, IX, and X	1983
Factor IIa inhibitor (Dabigatran)	Idarucizumab	Reducing unbound dabigatran concentrations with a median maximum reversal of 100% and a median time to bleeding cessation of 11.4 h	2015
Factor Xa inhibitors (e.g., rivaroxaban and apixaban)	Andexanet	Binding to factor Xa inhibitors, anti-factor Xa activity $\geq 90\%$, with reduced concentration of unbound direct factor Xa inhibitor and inhibited thrombin generation	2018
Factor Xa inhibitors and factor IIa inhibitors	Ciraparantag	Reversal of anticoagulation with factor Xa inhibitors and certain factor IIa inhibitors, exerts its effect through hydrogen bonding	Under research
Factor XI inhibitors	Asundexian	Near-complete factor XIa inhibition and inhibition of bidirectional cross between thrombin production and contact activation	Under research

nurses each bring their unique expertise to the table to form a dynamic and efficient collaboration system. Doctors take the lead in diagnosing and creating personalized anticoagulation plans. Pharmacists could stay updated on the latest drug information to assist doctors and nurses on drug interactions and potential adverse reactions^[65,66]. The FAERS data underscore the importance of educating patients about the risks and signs of bleeding, empowering them to seek timely medical attention to ensure the safe and effective use of anticoagulant drugs. When there is life-threatening bleeding or uncontrolled bleeding during initial resuscitation, anticoagulant reversal should be considered, and the use of this drug should be avoided or switched to other drugs^[67]. The present study provides the bleeding signal atlas of anticoagulant agents; prospective work is required to assess whether nurses or multidisciplinary medical teams-led protocols built on these signals reduce bleeding incidence.

Closing the quantitative–qualitative loop: from FAERS signals to multidisciplinary action

To convert the pharmacovigilance findings into front-line practice, we mapped the highest FAERS bleeding signals (Table 2) onto potential intervention directions. For patients with spinal subdural hemorrhage, nurses should conduct daily motor/sensory screening, pharmacists should check for potential drug interactions such as discontinuing CYP3A4/P-gp inhibitors, and doctors should perform spinal imaging as soon as possible and consider using dexamethasone $\pm$ neurosurgical decompression. Hemorrhagic arteriovenous malformation prompts strict blood-pressure control (<130/80 mmHg) and seizure precautions by nurses, and CT-angiography referral with possible embolization by physicians. Adrenal hemorrhage can be countered by nursing documentation of orthostatic vitals and flank-pain scores, and physician-ordered 08:00 cortisol/ACTH tests with endocrine consultation. Mesenteric hemorrhage requires nil-by-mouth status, large-bore i.v. access and cross-matching of 4 units, and CT-angiography $\pm$ interventional radiology embolization. Basal ganglia hemorrhage requires frequent neurological observations, control of systolic blood pressure < 140 mmHg; if it was related to warfarin, neurosurgical examination and INR reversal should be performed. Bullous hemorrhagic dermatosis is managed with non-adhesive dressings and lesion measurement, DOAC dose reconciliation against weight/

age, and skin biopsy or switch to LMWH if diagnostic uncertainty persists.

Lower-GI hemorrhage involves fecal occult-blood testing with serial hemoglobin monitoring, initiation or intensification of PPI, and early colonoscopy plus consideration of apixaban 2.5 mg bid as an alternative anticoagulant. Pericardial hemorrhage entails cardiac-tamponade surveillance and bedside echocardiography with drainage for effusion > 2 cm. Retroperitoneal hemorrhage requires inspection for flank bruising and hemoglobin checks, and CT quantification of bleeding with angiographic embolization if indicated. Finally, diverticular intestinal hemorrhage is addressed by a clear-fluid diet and stool-color charting, discontinuation of NSAIDs/antiplatelets, and CT-colonography or endoscopy. Embedding these signal-specific, role-assigned protocols in electronic health records converts FAERS disproportionality metrics into immediate bedside behavior, thereby closing the quantitative–qualitative loop and sustaining prospective risk surveillance.

Translating FAERS disproportionality into bedside action requires a drug-specific, anatomy-aware playbook. This study reveals that no anticoagulant is anatomically “neutral.” Oral agents uniformly generate high gastrointestinal signals, heparins and parenteral direct thrombin inhibitors cluster in the retroperitoneum, while betrixaban produces extreme but case-sparse RORs for traumatic and intracranial hemorrhage. Embedding these drug-specific PT signatures into clinical monitoring allows pharmacists and nurses to implement organ-focused surveillance, and physicians to pre-empt the dominant bleeding niche before it declares itself as a catastrophic event, thereby converting spontaneous-report statistics into real-time, multidisciplinary risk suppression.

Ongoing development of anticoagulant drugs with minimal bleeding risk

In recent years, the development of anticoagulant drugs with minimal bleeding risk has emerged as a key research focus (Table 5). Elucidating the intrinsic pathways of the coagulation cascade that are implicated in pathological thrombosis rather than hemostasis offers novel perspectives for innovative drug development^[68]. For instance, fucosylated glycosaminoglycan and its oligosaccharides have demonstrated anticoagulant and inhibitory effects on the intrinsic factor tenase complex, with

minimal risk of severe bleeding^[69]. Additionally, inhibiting factor XI or XIa has been shown to be more effective than low-molecular-weight heparin in preventing VTE following total knee replacement^[20]. Furthermore, patients with heart disease who take factor XI or XIa inhibitors exhibit lower bleeding risks compared to those on non-VKA OACs^[20].

Several compounds targeting factor XI, including Ionis-FXIRx, milvexian, asundexian, abelacimab, osocimab, and Xisomab 3G, have been developed, though their clinical benefits remain unsatisfactory to date^[68]. Encouragingly, the development of factor XI/XIa inhibitors holds promise for improving anticoagulation precision by selectively targeting thrombosis while preserving the patient's ability to clot normally, thereby reducing bleeding risks^[70]. Notably, factor XI inhibition appears particularly beneficial for patients with end-stage kidney disease or cancer, populations for whom NOAC use lacks clear indications^[71]. Further clinical validation is essential to confirm the efficacy and bleeding risk of these novel anticoagulant agents.

Limitations of this work

Our study has several limitations. The FAERS database, as a passive monitoring system, may have reporting bias, with factors like public awareness and media attention potentially causing over- or underreporting of bleeding events. Despite data cleaning efforts, FAERS data might still be incomplete or inaccurate, such as missing patient weight data, which could restrict our analysis of bleeding risk factors. When determining the causal link between anticoagulants and bleeding events, information bias may arise due to insufficient report details, making it hard to exclude other confounding factors. Though we used four algorithms for dispersion analysis, each has limitations, like ROR and PRR being sensitive to high values and less specific, while BCPNN and EBGM are more stable but may miss some signals. Our analysis also did not consider bleeding event classifications.

And due to data limitations, we could not evaluate and control all factors affecting bleeding risk, such as comorbidities and lifestyle factors (alcohol consumption, smoking, etc.). Furthermore, the existing studies lack quantitative research on the collaboration between nurses and other professionals. For example, there is a lack of quantitative data on the communication frequency and content between nurses and doctors during the formulation of anticoagulation treatment plans, as well as the degree of influence of nurses' opinions on doctors' decisions. There is insufficient research on nurses in the management of anticoagulation therapy for special populations (elderly and pediatric patients). There is also a lack of randomized controlled trials or high-quality cohort studies on the specific improvement degrees of nurses' intervention measures on indicators such as bleeding risk and the incidence of thrombotic events in patients. These are also the hot clinical issues that need to be addressed in the future research. Due to the rate of missing indications in FAERS reports and the wide variety of combination drug combinations, if stratified or interaction models are adopted, it will lead to insufficient sample sizes in a large number of cells and unstable signal estimation. Therefore, this paper did not conduct a quantitative assessment of the interaction between anticoagulants and specific concomitant medications. In the future, it is necessary to further clarify the additive or multiplicative

interaction effects of different anticoagulants with common concomitant medications such as antiplatelets, NSAIDs, and CYP3A4/P-gp inhibitors, with the help of an active monitoring platform.

Future insights to minimize anticoagulant-related bleeding

The administration of anticoagulant drugs has always been to minimize thrombus and bleeding, and there are many research directions that need to be overcome. First, real-world trials regarding anticoagulant drug interactions should be carried out, and novel strategies like dose adjustment based on therapeutic drug monitoring ought to be explored. However, dose adjustment grounded on concentration measurements is not advisable^[26]. Second, given the paucity of research on patients with hepatic insufficiency, more clinical trials are needed to enhance the safety of anticoagulant drugs in those with liver injury^[72]. Third, there is a need to assess the clinical outcomes of combining anticoagulant and antiplatelet therapies, particularly when used with novel anticoagulants, and to incorporate genotypes and phenotypes into anticoagulant decision making^[73]. Fourth, some studies have indicated that combining selective serotonin reuptake inhibitors with warfarin or dabigatran may raise the risk of bleeding, so it is important to find out the reasons behind this and develop suitable combination therapy plans^[26]. In new anticoagulant drug development and clinical practice, research into drug-drug interactions will help determine the safety of various drug combinations^[74].

Fifth, using multivariate analysis of variance and machine-learning studies can identify factors influencing bleeding and its severity, aiding in anticoagulant drug use assessment and promoting rational use. Sixth, future evaluations should leverage big data sources like FAERS to analyze bleeding events using comprehensive clinical data, thereby enhancing real-world efficacy and developing tools to mitigate or prevent bleeding. Future prospective studies need to link our high-signal PTs with laboratory-measured bleeding volumes to quantify the severity of bleeding. In future research, we also plan to attempt to better control confounding factors through multicenter and prospective studies, thereby improving the internal validity and external validity of the research. Seventh, the role of nurses in anticoagulation management is mainly based on literature reviews. Lastly, developing factor XI and XII inhibitors and nano-targeted agents could have specific anticoagulant effects and lower bleeding risks^[75,76]. There is also a pressing need to enhance existing systems and policies through targeted research, particularly to support and sustain nurse-pharmacist collaborations in clinical settings such as assisted reproduction^[77]. Future research is critically needed to discover new pharmaceutical care tools for minimizing DRP linked to anticoagulant drugs and to assess the clinical impact of nurse-pharmacist involvement in managing patients undergoing anticoagulant therapy. Clinical nurses constitute a crucial frontline contingent within the healthcare workforce. However, the studies identified few reports from registered nurses, and the nurses' knowledge of anticoagulants seems inadequate, and the academic degree positively influenced the extent of knowledge improvement^[78,79]. To address these deficiencies, implementing evidence-based training programs for both nursing students and practicing nurses is crucial, with a focus on the pharmacology, side effects, and drug-food interactions of high-risk anticoagulants like warfarin.

Additionally, encouraging postgraduate education and continuous professional development opportunities will further enhance nurses' knowledge and expertise^[54]. A recent study showed that nurse–pharmacist collaborations in community settings significantly enhance disease management and medication safety, reducing adverse drug events and hospitalizations while improving access to care and empowering patients with better chronic disease self-management skills^[77].

Conclusion

This study analyzed anticoagulant-related bleeding events using FAERS data, revealing significant variations in bleeding risks among different anticoagulants and emphasizing the vulnerability of elderly patients. The FAERS-generated signal atlas serves as an evidence base for designing multidisciplinary risk-management initiatives. Future research should also be directed toward investigating novel anticoagulant agents with improved safety profiles and exploring clinical practices to enhance the safety and effectiveness of anticoagulation therapy.

Ethical approval

This study utilized data from the FAERS database, a publicly available and anonymized resource designed to support the FDA-approved drug safety surveillance program and accessible to the public for research purposes. Data extraction and use were conducted in full compliance with the FAERS Data Use Agreement, including the FDA's Terms of Use for research purposes. Given the anonymized nature of the data and its intended use for surveillance and research, Institutional Review Board approval, as well as patient consent, were not required.

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Author contributions

YZW and XH: Conceptualization. KLM and LCL: Data curation. QW and ZLW: Formal analysis. KLM, LCL, and XH:

Investigation. KLM and LCL: Methodology. YZW and XH: Project administration. XH and KLM: Resources. QW: Supervision. XH, KLM, and LCL: Writing – original draft. XH, QW, ZLW, LCL, KLM, and YZW: Writing – review & editing.

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All authors declare no competing interests.

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Data are available in a public, open-access repository. Data are available on reasonable request. All data relevant to the study are included in the article or uploaded as supplementary information.

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